

The External Cortex Loop: Contingency Interruption, Reinstatement, and Cognitive Incorporation in EEG Neurofeedback

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Preprint Version 1.0 - 1 August 2026

Abstract

A display that happens to accompany brain activity is not yet part of a cognitive loop. A more demanding possibility arises when current brain state changes an external sensory signal, the person learns the contingency, and the returned signal begins to guide perception, regulation, or action. This paper asks when such a loop becomes functionally incorporated and how incorporation can be distinguished from novelty, expectancy, ordinary sensory stimulation, and device failure. The proposal originates in the Neo Mind Cycle program. Public 2012 material described EEG-driven light and sound as a recursive route by which a learning mind could observe transformed aspects of its own activity; a 2017 Neural Lace Podcast developed bidirectional neural communication and feedback/no-feedback protocols; and 2021 Recorder files preserve a first-person report that the feedback felt like an extension into a computer and that its unexpected cessation felt like falling out of an airplane. That report is a hypothesis generator, not efficacy evidence. The paper defines a stricter test: participants learn a person-specific EEG-to-audiovisual mapping; the causal contingency is then concealedly interrupted while sensory statistics are preserved; the exact learned mapping is reinstated; and a mismatched mapping is tested as a specificity control. Twenty equations formalize neural features, device transformation, co-adaptation, conditional contingency, acquisition, interruption, reinstatement, mapping identity, transfer, mediation, and a conjunctive incorporation gate. The primary endpoint is a preregistered task-specific behavioral effect. Agency, ownership, and phenomenology are secondary; EEG change is mechanistic; all are separated from therapeutic claims. Incorporation requires acquisition above replay, selective loss when contingency is broken, selective recovery when the same mapping returns, failure of a mismatched mapping to rescue, and controlled latency and artifact results. This interruption-and-reinstatement design is narrower than a general extended-mind thesis and adds a causal assay for determining when an external neurofeedback process has become part of an operative human-computer control loop. Keywords: EEG neurofeedback; external cortex; brain-computer interface; closed loop; contingency; co-adaptation; agency; embodiment; interruption; reinstatement; Self Aware Networks

1. The ordinary experience and the scientific problem

Imagine wearing a system that turns features of your current EEG into changing light and sound. At first, the display is merely unusual. With practice, some changes become predictable: a shift in attention or arousal is followed by a corresponding change in the returned signal. You may begin to act with the loop rather than simply watch it.

Now imagine that the display continues, with the same brightness, volume, motion, and update rate, but it is no longer driven by your current brain state. If nothing changes in performance or experience, the loop may never have become functionally important. If a task-specific ability selectively degrades, and if the ability returns only when the exact learned mapping is restored, the evidence is stronger. The causal contingency, rather than the presence of stimulation alone, has become part of the operative system.

This is the problem addressed here. The paper does not ask whether tools can be useful in general. It does not ask whether a computer literally becomes biological cortex. It asks for a measurable criterion by which an external, brain-contingent sensory process can be said to have been incorporated into a cognitive operation.

Three confounds make the question difficult. First, a broken display is obvious and creates surprise. Second, neurofeedback effects can arise from expectation, coaching, repetition, reward, and sensory stimulation rather than the targeted brain signal [14,15]. Third, human and decoder can change together, so a fixed pre/post comparison can miss the dynamics of co-adaptation [10-13]. A credible experiment must preserve sensory appearance during interruption, identify what each learner adapts, and demand a mapping-specific rescue.

2. From instrument to incorporated loop

2.1 Five architectures

The word "interface" covers systems with different causal structures.

1. Ordinary tool: the person acts through muscles; the device returns ordinary sensory consequences.
2. Decoder: neural activity is mapped to an output, but the returned signal need not change the user's future neural strategy.
3. Feedback display: a neural measure changes a display that the user can observe.
4. Reciprocal adaptive loop: the person changes in response to feedback while the decoder or mapping also changes.
5. Incorporated contingent loop: a learned operation becomes selectively dependent on the current-person contingency and can be disrupted and rescued by manipulating that contingency.

The first four do not automatically establish the fifth. High classification accuracy can coexist with low incorporation. A compelling subjective report can coexist with unchanged task performance. Conversely, useful control can be learned without a strong feeling of ownership.

2.2 Functional incorporation, agency, and ownership

Functional incorporation means that the external process participates causally in a defined operation. Agency is the experience or judgment that one controls an event. Ownership is the experience or judgment that an entity or state belongs to oneself [18]. These variables may covary but must be measured separately. A loop can improve performance without being experienced as part of the self, or produce a vivid ownership illusion without improving performance [16,17].

2.3 The term "external cortex"

The operation should be understood before it is named. A person's changing neural state is measured. A transformation converts selected features into a sensory signal. The returned signal changes the person's next state. With repeated reciprocal exchange, the person learns how actions or internal strategies alter the loop. If the learned contingency becomes necessary for a task and its exact return restores the task, the external transformation has joined the operative control circuit.

External cortex loop is the proposed name for that reciprocal, learned, causally incorporated architecture. The term is functional, not anatomical. It does not assert that a monitor is histologically cortex, that cognition has migrated out of the skull, or that all feedback systems are parts of the self.

3. Development of the proposal, 2011-2026

Stage	Source-bound contribution	Evidentiary boundary
2011-2012 apparatus period	Neo Mind Cycle combined EEG with contingent light and sound; later source notes distinguish mirrored brain-state feedback from rigid command decoding [1-3].	Some apparatus dates are internal or retrospective. The strongest public anchors are the 2012 videos, not every claimed 2011 experiment date.

Stage	Source-bound contribution	Evidentiary boundary
January-August 2012	The term Self Aware Networks appears in a private Facebook export, while public videos describe recursive EEG feedback, self-observation, prediction, and a mind-computer loop [1,2,4].	The export proves group membership and name only. The public videos establish the recursive concept but not modern efficacy.
April 2017	The Neural Lace Podcast asks when neural communication should use feedback, develops read/write BCI architecture, and distinguishes handshake-like from non-handshake transmission [5].	This is a public operational ancestor. It does not contain the present interruption/reinstatement experiment.
September-October 2021	Recorder notes describe the loop as an extension of awareness into a computer. One recording reports immediately detecting that EEG-driven feedback had stopped and compares the loss to falling out of an airplane [6,7].	The audio files carry 2021 creation metadata. The claimed 2011 experience is retrospective, and the phenomenology is not a controlled result.
June 2022 onward	Public SAN Git material integrates receive-transform-project-feedback, neural rendering, feedback, sensory substitution, and artificial-neurology branches [8].	Mature SAN terminology is publicly fixed in 2022; it is not backdated to 2011.
July 2026	The final-preprint Thought-Density Hypertime and the Extended Brain defines a broad reciprocal-external-cortex architecture, matched replay, latency, provenance, identity, decoupling, and a preparticipant protocol [9].	That paper owns the broad extension model. The present paper must add a narrower causal assay rather than duplicate it.
August 2026	The current paper specifies concealed contingency interruption, exact reinstatement, mismatched rescue, and a conjunctive incorporation criterion.	These formal measures and experiments are current contributions.

The 2021 "airplane" report is scientifically useful because it nominates a precise contrast: expected brain-contingent return versus the sudden absence of that return. It cannot establish incorporation on its own. Device failure may be salient, the sensory stream may visibly stop, and recollection may be shaped by the unusual experience. The correct response is not to discard the report or promote it to proof. It is to turn the reported contrast into a blinded experiment.

4. Established evidence and the residual question

4.1 Closed-loop neurofeedback

Neurofeedback provides information about measured brain activity so that participants can learn to alter a target signal [14]. The field has demonstrated learning in multiple modalities, but results depend on target definition, feedback design, strategy, controls, and reporting quality. CRED-nf therefore requires explicit control groups, feedback specifications, brain and behavioral outcomes, and data practices [15]. These standards are adopted here.

Neurofeedback does not automatically reveal what a brain variable "means" to the participant. A person may exploit any stable cue available. The present design therefore asks whether the current-person contingency matters after controlling the returned signal's sensory statistics.

4.2 Artificial sensory feedback and body updating

Artificial sensory signals can be learned and integrated with other information [22]. Body representations also change through temporally and spatially structured multisensory evidence and tool use [16,17]. These findings establish plasticity and incorporation-like phenomena. They do not show that a specific EEG-contingent audiovisual loop has become necessary for a task.

4.3 Co-adaptive interfaces

Closed-loop neural interfaces contain at least two learners: the user and the machine. Madduri and colleagues model how their learning rates and policies interact [10]. Recent noninvasive studies show that neural manifold geometry constrains learnability and that sensory-guided joint learning can improve acquisition [11,12]. A loop may fail because the user cannot reach decoder-relevant states, because the decoder moves too quickly, or because both adapt in incompatible directions. Interruption tests must therefore freeze or explicitly model the machine during causal evaluation.

4.4 Extended cognition and the missing discriminator

Licklider's symbiosis program and Clark and Chalmers' extended-mind argument motivate functional integration across organism and artifact [20,21]. Philosophical parity, reliability, accessibility, and coupling are useful criteria, but they do not provide a laboratory-specific rescue test. The residual question is:

Does a learned operation depend on the individual's live neural-to-sensory contingency, and is that dependence selectively restored by the same mapping?

That question is experimentally separable from whether an artifact should count as part of the person in a broader philosophical or legal sense.

5. The proposed operation before specialized terminology

A usable assay begins with a task rather than an acronym. The participant must hold a cursor steady, discriminate a hidden state, regulate an arousal-related measure, or integrate returned feedback with another sensory estimate. The system measures a preregistered EEG feature vector. A fixed transformation produces audiovisual feedback. Training continues until performance reaches a preregistered acquisition criterion.

The critical session then contains short, covertly randomized blocks:

- Contingent: the display is driven by the participant's current feature vector.
- Replay: the participant receives a previously recorded stream with matched sensory statistics.
- Phase-scrambled or conditional surrogate: local spectra and amplitudes are preserved while current-person temporal contingency is removed.
- Latency shift: the correct mapping is delayed by a controlled amount.
- Exact reinstatement: the original mapping and latency return.
- Mismatched reinstatement: a mapping learned by another participant, or an orthogonal transformation of the same features, is substituted.

The participant is not told when blocks change. Visual and auditory continuity, reward frequency, task difficulty, and experimenter contact are matched. A system-status marker is logged but not shown. The primary question is not whether the person notices something. It is whether task performance shows a selective causal signature.

Only after this operation is clear do we use the term external cortex loop. Incorporation is the empirical pattern, not the metaphor.

6. Formal model

Let $e_t \in \mathbb{R}^p$ be preprocessed EEG samples and let z_t be a preregistered feature vector:

$$z_t = \Phi(e_{t-W:t}), \quad (1)$$

where Φ may include band power, phase relations, spatial filters, or event-related features. The device maps current features and decoder parameters to a sensory state:

$$y_t = D_{\theta_t}(z_t). \quad (2)$$

The participant's state h_t changes as a function of task input, action, and returned signal:

$$h_{t+1} = H_{\psi_t}(h_t, s_t, a_t, y_t). \quad (3)$$

Co-adaptation permits both parameter sets to change:

$$\begin{aligned} \theta_{t+1} &= \theta_t + \eta_D \text{grad}_{\theta} J_D, \\ \psi_{t+1} &= \psi_t + \eta_H \text{grad}_{\psi} J_H. \end{aligned} \quad (4)$$

During causal test blocks, θ is frozen unless decoder adaptation itself is the preregistered variable. Live contingency can be quantified as conditional mutual information:

$$C_{\text{live}} = I(z_t, y_{t+\delta} | y_{t-1:t-k}, s_t). \quad (5)$$

A replay or surrogate should preserve marginal sensory statistics while reducing current-person contingency:

$$\begin{aligned} p(y | B_{\text{cont}}) &\approx p(y | B_{\text{sur}}), \\ C_{\text{cont}} &> C_{\text{sur}}. \end{aligned} \quad (6)$$

Acquisition relative to matched replay is

$$A = P_{\text{late,cont}} - P_{\text{early,replay}}. \quad (7)$$

The causal interruption effect is

$$I = P_{\text{cont,pre}} - P_{\text{broken}}. \quad (8)$$

Exact reinstatement rescue is

$$R = P_{\text{exact restore}} - P_{\text{broken}}. \quad (9)$$

Mapping specificity requires exact rescue to exceed mismatched rescue:

$$S_R = R_{\text{exact}} - R_{\text{mismatch}}. \quad (10)$$

Latency sensitivity is estimated as a function of imposed delay:

$$L(\delta)=E[P|\delta]-E[P|\delta=0]. \quad (11)$$

Mapping identity can be measured by the distance between learned and test transformations:

$$d_D(\theta_a, \theta_b)=E_z[|D_{\theta_a}(z)-D_{\theta_b}(z)|_2]. \quad (12)$$

The predicted rescue curve decreases with mapping distance after sensory similarity is controlled:

$$\begin{aligned} R(d) &= \beta_0 - \beta_1 d_D + \varepsilon, \\ \beta_1 &> 0. \end{aligned} \quad (13)$$

Transfer to an untrained task u is

$$T_u = P_{u, cont} - P_{u, replay}. \quad (14)$$

Agency and ownership remain separate secondary variables:

$$Q_{agency} \neq Q_{ownership} \neq P_{task}. \quad (15)$$

A mediation model tests whether learned neural control explains behavioral benefit:

$$\Delta z_{control} = f(B_{condition}), \quad \Delta P_{task} = g(\Delta z_{control}). \quad (16)$$

An artifact gate excludes intervals in which feedback dependence is explained by eye, muscle, or movement signals:

$$G_{artifact} = I[\max(q_{EOG}, q_{EMG}, q_{move}) < q_*]. \quad (17)$$

Blind-state detection is estimated from participant guesses about block identity:

$$D_{blind} = AUC(B_{guess}, B). \quad (18)$$

If task effects remain after conditioning on detected block identity, demand is less likely to explain the result:

$$\tau_{cont} = E[P|B_{cont}, B_{guess}] - E[P|B_{sur}, B_{guess}]. \quad (19)$$

The incorporation gate is conjunctive:

$$E_{inc} = AND(A^+, I^+, R_{exact}^+, S_R^+, G_{artifact}, G_{blind}). \quad (20)$$

A vivid report cannot substitute for a failed causal gate.

7. Experimental protocol

7.1 Stage 0: preregistration and feasibility

Specify the EEG hardware, electrode montage, reference, preprocessing, artifact rejection, feature family, update rate, total latency, audiovisual transformation, task, primary endpoint, stopping rules, exclusion rules, and smallest effect of interest. Run a separate engineering feasibility cohort to confirm that the system can maintain matched sensory statistics across contingent and surrogate modes. Do not use the feasibility cohort for confirmatory inference.

7.2 Stage 1: calibration without moving targets

Collect resting and task EEG, identify artifact-sensitive channels, and fit the initial mapping on a predefined training subset. The mapping should be interpretable enough to determine what exact reinstatement means. If a deep decoder is used, version and hash the model, feature scaler, weights, and runtime. A participant-specific mapping and a sensory-matched surrogate must be frozen before confirmatory blocks.

7.3 Stage 2: acquisition

Randomize participants to live-contingent training, yoked replay, or a sensory-matched noncontingent control. Keep instructions and reward identical. Continue for a preregistered number of sessions rather than stopping only successful learners. Model learning curves and report the fraction meeting the acquisition criterion.

Acquisition requires more than improved classifier output. The participant must improve a task that uses the returned signal, and improvement must exceed replay or control. EEG feature control, task performance, and subjective strategy are reported separately.

7.4 Stage 3: concealed interruption

Within participants who complete training, interleave contingent, replay, conditional-surrogate, latency, and blank-failure blocks. Blank failure is included to estimate the effect of an obvious device outage; it is not the critical control. The conditional surrogate should preserve brightness, color distribution, sound level, motion energy, update rate, and reward schedule while destroying dependence on current EEG.

Use short blocks and a cover task to reduce explicit strategy changes. Record block-identity guesses after only a subset of blocks so the query itself does not reveal the manipulation.

7.5 Stage 4: exact and mismatched reinstatement

After a broken-contingency block, restore the original model, feature scaler, latency, and sensory mapping. In separate blocks, restore a mapping with similar output statistics but different participant identity or feature geometry. Exact rescue without mismatched rescue is the key discriminator. If any plausible contingent mapping rescues performance, the result supports live feedback in general rather than incorporation of a particular learned mapping.

7.6 Stage 5: transfer, persistence, and washout

Test an untrained task that can use the same signal, then remove the device for a washout interval. Reassess whether performance returns to baseline, whether the learned internal strategy persists, and whether reacquisition is faster. Persistence without the device may reflect internal learning rather than continued external dependence; that is a useful outcome but not evidence that the absent device remains part of the active loop.

8. Analysis, controls, and identifiability

8.1 Primary endpoint

Use one behavioral primary endpoint tied to the trained operation. Examples include tracking error, discrimination accuracy, regulation success under a held-out challenge, or response-time/accuracy tradeoff. Select it before recruitment. Agency, ownership, vividness, and "extension" ratings are secondary.

8.2 Hierarchical model

Estimate participant-level acquisition, interruption, and rescue within a multilevel model containing condition, session, block, mapping distance, latency, and their preregistered interactions. Include random participant intercepts and slopes where supported. Report raw trajectories as well as model estimates.

8.3 Sensory-equivalence audit

For every block type, compare luminance, color histogram, acoustic envelope, spectral distribution, motion energy, event rate, reward rate, and transition frequency. Failure to match these variables converts a contingency test into a stimulus test. Publish the audit and the surrogate-generation code.

8.4 Artifact and strategy audit

Demonstrate that eye movement, facial muscle, jaw tension, posture, and ordinary controller movement do not carry the decisive mapping. Re-run analyses after removing contaminated intervals and after regressing nuisance channels. Report participant strategies verbatim in a de-identified supplement and test whether one overt strategy explains the effect.

8.5 Decoder drift audit

During confirmatory interruption and rescue, either freeze the decoder or version every update. If the decoder continues to adapt, the experiment no longer knows whether loss or rescue arose from the participant, the machine, or both. Co-adaptation can be a separate preregistered phase after the fixed-mapping causal assay.

8.6 Expectancy and blinding

Measure guessed condition, perceived device quality, expected benefit, and experimenter contact. A fully double-blind interface may be difficult, but allocation, block scheduling, and primary analysis can be automated. The analyst should remain blind to condition labels until the pipeline is frozen.

8.7 Multiplicity and missingness

Control the family of secondary subjective outcomes. Report all preregistered conditions, participants, adverse events, and exclusions. Analyze attrition under plausible missing-data models because participants who fail to gain control may be more likely to withdraw.

9. Predictions and failure conditions

The strongest predicted pattern is:

1. live-contingent training exceeds matched replay during acquisition;
2. concealed contingency interruption selectively impairs the trained operation;
3. exact reinstatement restores it;
4. a sensory-matched but mapping-mismatched reinstatement does not fully restore it;
5. increasing latency or mapping distance produces an ordered loss;

6. the pattern survives artifact removal and cannot be explained by guessed condition;
7. subjective agency or ownership may track the effect but is not required to equal it.

The following results would weaken or reject incorporation:

- Stimulation-only result: live, replay, and surrogate feedback perform equally.
- Obvious-outage result: impairment occurs only when the display visibly stops.
- Generic-contingency result: any live mapping rescues equally well.
- Expectancy result: performance follows guessed condition rather than actual contingency.
- Artifact result: eye or muscle activity explains the decoder and the behavioral effect.
- No-acquisition result: participants never outperform replay before interruption.
- Subjective-only result: extension ratings change while the preregistered task does not.
- Internalization result: training improves later unaided performance, but interruption no longer matters. This supports learning, not current external dependence.

The last outcome is not a failure of neurofeedback. It simply answers a different question: the loop trained an internal skill rather than remaining an incorporated online component.

10. Distinction from related SAN and extended-brain claims

The broad Thought-Density Hypertime and the Extended Brain manuscript asks how reciprocal external systems may expand cognitive throughput, event density, memory, and subjective time [9]. It already defines ordinary tools, decoders, reciprocal cortex, autonomous artificial brains, provenance, latency, identity, and decoupling. The present paper does not repeat that architecture.

Its distinct contribution is the causal signature of incorporation: acquire, conceal, interrupt, exactly reinstate, mismatch, and test rescue. This assay can be used even by researchers who reject extended-mind language. It also places a stricter burden on the 2021 first-person observation. The reported airplane-like loss motivates the interruption manipulation, but the experiment succeeds only if a blinded, sensory-matched, task-specific, mapping-specific effect appears.

The proposal also fits SAN's receive-transform-project-feedback architecture. EEG is a coarse measurement of population electrical activity, the device transforms selected features, the audiovisual display projects a new sensory pattern, and the participant's sensory systems receive it. The next neural state changes the next measured signal. This is a literal feedback loop. Whether the loop participates in consciousness, selfhood, or only task control remains an empirical question and should not be assumed from reciprocity alone.

11. Safety, privacy, and interpretive limits

Noninvasive EEG does not make the experiment risk-free. Light and sound must remain within safety limits; flicker frequencies and participant seizure history require appropriate screening and ethics review. Participants should be able to pause or withdraw immediately. The study must not promise therapy, cognitive enhancement, life extension, or machine-mediated consciousness.

Neural data are identifying and sensitive. Store raw EEG, event logs, mappings, and subjective reports under a data-management plan that separates participant identity from public materials. Public reproducibility can use de-identified derivatives, synthetic fixtures, source code, and cryptographic hashes without releasing private raw recordings by default.

The term external cortex may invite metaphysical or marketing inflation. A passed incorporation gate establishes participation in one defined operation over one time interval. It does not prove permanent identity integration, moral personhood of the device, transfer of consciousness, or anatomical equivalence to cortex.

Historical provenance must also remain layered. The public 2012 videos are stronger public anchors than later recollection. The 2021 audio file is a dated source for the report made in 2021, not automatic proof that the described experience occurred in 2011. The 2017 podcast establishes a public feedback-protocol ancestry but not the present randomized design. The 2026 experiment and equations retain their actual date.

12. Conclusion

An external signal becomes scientifically interesting when it does more than mirror a measurement. It becomes part of a candidate cognitive loop when the person learns its current-state contingency, a task comes to depend on that contingency, covert interruption selectively impairs the task, and exact reinstatement selectively restores it.

The external cortex loop is therefore defined by a causal test, not by proximity to the body or by the vividness of a metaphor. The proposal turns a 2011-2021 Neo Mind Cycle experience into a modern preregisterable experiment while preserving its evidentiary limits. If the conjunctive gate passes, the result would show that a person-specific EEG-to-sensory transformation can become an operative component of a learned human-computer system. If it fails, the design will identify whether the failure lies in acquisition, contingency, mapping specificity, latency, artifact control, or subjective interpretation.

Author contributions and AI-assisted development

Micah Blumberg conceived Neo Mind Cycle, the external-cortex hypothesis, the historical first-person observations, and the interruption/reinstatement research question. AI tools assisted with source routing, bibliographic verification, mathematical formalization, adversarial controls, and editorial production under the author's direction. All claims, interpretations, and publication decisions remain the author's responsibility.

Data and materials availability

This is a theory-and-methods paper. No participant data were collected. The final preprint package contains a source manifest, claim-source-genealogy atlas, figure plan, bibliography, and validation receipt. The historical private files remain under author custody; this package does not authorize their public release.

Ethics statement

No human-subject experiment was performed for this manuscript. Future implementation requires institutional ethics review, informed consent, flicker and seizure-safety assessment, adverse-event monitoring, neural-data privacy safeguards, preregistration, and a participant withdrawal procedure.

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